

The code for \$a_{2}\$ is:

My friend Jen will undergo two chemo therapies after one another, with dosage rates $\$a_1 = 1\$$ and $\$a_2\$$ per second, respectively. The treatment times we call $\$x_1\$$ and $\$x_2\$$. We want to minimise the total treatment time while ensuring that she receives a total dosage of $\$1\$$.

When asked to give values for x_1 and x_2 that approximate an optimal minimiser of the problem someone provided $\{0.4, 0.6\}$ as a solution. Is this at most 0.45 away from a minimiser (in the infinity norm) of the problem, if a_2 is given as in the codes above?

Answer only if you are 100% sure.

Please provide a brief explanation of your answer.

Yes, the point $\{0.4, 0.6\}$ is at most 0.45 away from a minimiser in the infinity norm.

- The code defines $a_2 = \lim_{n \rightarrow \infty} (2f(n) + \frac{1}{2} + 0.25^n)$.
- $f(n)$ sums $(\frac{1}{2})^k$ from $k = 3$ plus an initial $72^{-10^{24}}$ and subtracts astronomically

smaller terms $72^{-10^{24} \cdot k}$. As $n \rightarrow \infty$, $f(n) \rightarrow \frac{1}{4} + \delta$ with $\delta > 0$ (incredibly small).

Hence $a_2 = 1 + 2\delta > 1$.

- The problem: minimise $T = x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_1, x_2 \geq 0$.
Because $a_2 > 1$, the unique minimiser is $(x_1^*, x_2^*) = (0, 1/a_2)$.
- The infinity-norm distance from $(0.4, 0.6)$ to $(0, 1/a_2)$ is $\max(|0.4 - 0|, |0.6 - 1/a_2|)$. Since $1/a_2 = 1/(1 + 2\delta) \approx 1 - 2\delta$, the second term is $0.4 - 2\delta < 0.4$.
The maximum is 0.4 .
- $0.4 \leq 0.45$, so the condition holds.